T-LEVELS | Institute for Apprenticeships & Technical Education

Please check the examination details below before entering your candidate information

| Candidate surname | Other names |
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Centre Number

Learner ID LL-

**T Level Technical Qualification in Digital Production, Design and Development (Level 3)**

**Time** 2 hours 30 minutes

**Paper reference** **19537**

# Core

## PAPER 2: The Business Environment

**You do not need any other materials.**

Total Marks

## Instructions

- Use **black** ink or a ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and Pearson learner ID.
- There are two sections in this question paper. Answer **all** questions in Section A and Section B.
- Answer the questions in the spaces provided – *there may be more space than you need.*

## Information

- The total mark for this paper is 100.
- The marks for **each** question are shown in brackets – *use this as a guide to how much time to spend on each question.*

## Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

*Turn over*


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## SECTION A

**Answer ALL questions. Write your answers in the spaces provided.**

**1** (a) Explain what is meant by a business that provides a product.

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(b) Name **two** layers of the 4 layer TCP/IP model.

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**(Total for Question 1 = 4 marks)**

**2** (a) Name **two** items of information that a company would wish to keep private and confidential.

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(b) Explain **two technical** threats to digital systems.

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**(Total for Question 2 = 6 marks)**

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**3** (a) Describe how a device may be connected to the internet of things (IoT).

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(b) Explain **two** characteristics of Random Access Memory (RAM).

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(c) Describe how data may be transformed into information.

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**(Total for Question 3 = 10 marks)**

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**4** A cyber security company is developing a new anti-virus tool.

Explain **two** reasons why the company would use virtual environments to develop and test this new tool.

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**(Total for Question 4 = 4 marks)**

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**5** *SuperShoes* designs and creates leisure footwear.

Explain **two** ways *SuperShoes* could use digital tools in the design of new products.

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**(Total for Question 5 = 4 marks)**

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**6** Explain **one** potential risk to an organisation of the use of digital technology.

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**(Total for Question 6 = 3 marks)**

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**7** Evaluate the use of Data Analysis Tools and their use in business.

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**(Total for Question 7 = 9 marks)**

**TOTAL FOR SECTION A = 40 MARKS**

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## SECTION B

**Answer ALL questions. Write your answers in the spaces provided.**

**8** A supermarket wants to upgrade its digital systems.

It wishes to move away from a desktop application-based system to a system that uses a cloud environment.

(a) Explain **two** ways that this new system will be able to support the key areas of organisation in the supermarket.

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(b) Describe how a pilot changeover strategy could be used.

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(c) Explain **one** way that the new cloud-based system could meet user needs.

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(d) Discuss how Data as a Service (DaaS) could be used by the supermarket.

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**(Total for Question 8 = 15 marks)**

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**9** A leisure centre wishes to install a new digitised management system.

Functions of the new system include storing member details and allowing members to pay for fitness classes.

(a) The change was triggered by a wish to add new features to improve the customer experience.

Describe **two other** factors that could trigger change.

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(b) Explain **two** potential benefits to leisure centre stakeholders that this new system could provide.

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(c) Explain why it is important to ensure that this new system is resilient.

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(d) Evaluate the risk of **human threats** to the new leisure centre system.

Your evaluation should include:

- possible threats to the system
- the severity of each threat
- the likelihood of each threat
- a conclusion as to which threat would be of most concern to the leisure centre.

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**10** A retailer is developing a new online shopping system.

Customers can log into their accounts to search for items and place orders.

(a) Describe **two ways** in which data may be generated by the new online shopping system.

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(b) A conceptual data model has been designed. The model contains these entities:

- Customer.
- Item.

Describe how these entities could be linked together.

**(3)**

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(c) When a customer searches for an item, the server sends information about the product back to the client as a JSON file.

Explain why a JSON file is used in this situation.

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(d) The new system will use RAID 1.

Describe how RAID 1 stores data.

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(e) The developers of the system wish to use multi-factor authentication methods to secure user accounts.

They must choose between either possession-based systems or biometric systems.

Evaluate the suitability of using these methods to secure user accounts.

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**(Total for Question 10 = 25 marks)**

**TOTAL FOR SECTION B = 60 MARKS**
**TOTAL FOR PAPER = 100 MARKS**

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